

Prashant Sah

[LinkedIn](#) | [Website](#) | prashantsah2061@gmail.com | [GitHub](#)

Education

Caldwell University
Bachelors in Computer Science

Caldwell, NJ
May 2027

- Dean's List, Honors Award Recipient, Presidential Scholarship
- Courses: Data Structures & Algorithms, Web Development, Database Management, Calculus, Artificial Intelligence

Projects

Video Content RAG System

[GitHub](#)

- Engineered end-to-end machine learning pipelines using Python, pandas, and scikit-learn for automated data preprocessing, feature extraction, and model training.
- Implemented natural language processing (NLP) workflows integrating OpenAI Whisper and BGE-M3 embeddings for semantic text analysis and context understanding.
- Developed retrieval-augmented generation (RAG) models leveraging vector embeddings and cosine similarity for context-aware information retrieval.
- Automated data ingestion, transformation, and parallel processing for JSON and multimedia datasets using pandas, NumPy, and Joblib to optimize pipeline efficiency.

California Housing Price Predictor

[GitHub](#)

- Built a machine learning web app using Flask and Random Forest Regressor to perform data-driven house price prediction across California.
- Performed data analysis and manipulation using pandas and NumPy to preprocess, clean, and transform large housing datasets.
- Implemented automated preprocessing pipelines with feature engineering and stratified cross-validation for improved model accuracy.
- Designed an interactive interface with Bootstrap 5 supporting both real-time and batch predictions via CSV uploads.

Experience

Caldwell University, Teaching Assistant – CS 196 & CS 302

Jan 2025 – Present

- Assisted in developing and grading Python programming assignments for 50+ students in CS 196, covering OOP, recursion, and data manipulation.
- Conducted lab sessions for 50+ students on data analysis and visualization using NumPy, Pandas, Matplotlib/Seaborn
- Supported instruction for 15 students in CS 302 on algorithm design, graph theory, and computational problem-solving.
- Reviewed and debugged student code to enhance efficiency, readability, and analytical accuracy.
- Developed materials to enhance understanding of programming logic, data structures, and data analysis.

Caldwell University, Student Tech – IT Department

July 2024 – Present

- Supported campus-wide AV and IT systems, ensuring reliable data and technology operations.
- Assisted faculty and students with troubleshooting university portal and system access issues.
- Rebuilt and configured computers, optimizing system performance and data processing efficiency.
- Collaborated with IT staff to maintain hardware, software, and network integrity for data-driven tasks.

Caldwell University, Peer Tutor – Academic Success Center

Sept 2024 – Present

- Tutored 25+ students in Python and algorithms, strengthening their computational problem-solving skills.
- Guided learners through topics in web development, discrete math, and algorithmic logic.
- Assisted students in applying programming concepts to data analysis and efficient code design.

Caldwell University, Barnes & Noble Campus Store Team Member

July 2024 – Present

- Managed textbook and inventory data to ensure accurate tracking and timely order fulfillment.
- Assisted in coordinating course materials by analyzing syllabus requirements and sales trends.
- Operated POS and digital inventory systems with attention to data accuracy and efficiency.

Skills

Languages: Python, JavaScript, C, SQL (MySQL), HTML/CSS

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib, Flask

Tools: Git/GitHub, Jupyter Notebook, VS Code, Excel, Word

Domains: Machine Learning, Data Analysis, NLP, Data Visualization